Protocol Sequences for Mobile Ad Hoc Networks

Yi Wu, Kenneth W. Shum, Zihuai Lin, Wing Shing Wong, and Lianfeng Shen

Abstract— **Protocol sequences offer a promising alternative for media access control of mobile ad hoc networks, because they do not require any coordination among the users nor any centralized synchronization. We show that by using suitably designed deterministic scheduling, the delay performance can indeed be much better than using random and pseudo-random sequences. The reported results indicate that protocol sequences can offer practical solutions to complicated multiple-access problems in ad hoc networks, such as vehicular ad hoc networks (VANET). The cumulative distribution function of delay and an upper bound of the individual delay in the cases of protocol sequences are derived.**

Keywords: Protocol sequences, CRT sequences, user-irrepressible sequences, mobile ad hoc network, slotted ALOHA.

I. Introduction

The lack of a centralized timing and scheduling is one of the sources for the challenges in the design of *mobile ad hoc networks* (MANET). One example is vehicle-to-vehicle communication systems that enable users disseminate information, e.g. speed and direction, to their neighbors. As users may join or leave the system at a fast rate, protocols such as electing one of the mobile users as a central access point using a leader election algorithm [1] is not always feasible.

Besides centralized control, there are two traditional contention-based protocols for channel access: carrier-sense multiple access (CSMA) [3] and ALOHA [4]. The CSMA protocol requires a backoff algorithm which is not easy to implement in a MANET and may not be suitable for quick dissemination of data in high-mobility system. For simplicity of system description, we will assume slot synchronization distributed protocol which operates in the absence of centralized scheduler is devised in this paper.

A simple channel-access protocol without centralized scheduler is slotted ALOHA [5]. In slotted ALOHA, each user sends a packet in a time slot with some fixed probability. Random numbers are required in the implementation. In practice, we use deterministic finite-state generators instead of truly random numbers. The deterministic binary sequences used for accessing the channel are called *protocol sequences* in [6]. The zeros and ones in a protocol sequence are read out periodically, and a packet is sent if and only if it is one. If the deterministic binary sequences look like random sequences statistically, we expect that the resulting system performance metrics, such as throughput and delay, are similar. In this paper, we propose using a special class of protocol sequences, which is statistically different from truly random sequences, so that the delay performance can be significantly improved.

In [7], analogous results for frame-synchronous system, with applications to multiple access in satellite networks, are reported. The authors in [7] assume that the time slots are indexed by a central coordinator (the satellite for instance), and each user knows the number of the current time slot, modulo the sequence period. In this paper, we only assume a slot-synchronous model, and no central coordinator and no global time-slot indexing is required. Besides the application in the area of MANET, the proposed protocol sequence can also be applied in some other areas. For example, it can be used in the area of Body Area Networks (BAN) [2] for the purpose of medical monitoring for instance.

II. System Model and Definitions

Consider a MANET consisting of M users, each of them is within the transmission range of each other and wants to broadcast data to everyone else. We assume that the users always have something to send. A time-slotted collision channel is used. In this paper, we assume that the system is slot-synchronous, meaning that at any receiver all received packets are contained in one slot. At any time slot, there is a collision if there is more than one transmitted packet. In such a case, all transmitted packets are considered lost.

In a deterministic channel access scheme, user i is statically assigned a periodic binary (0/1) sequence, $s_i(t)$, of period L. User i starts transmitting at τ_i, which is called the *relative delay offset*, and transmits a packet at time slot $t+\tau_i$ if $s_i(t) = 1$, but listens to incoming packet if $s_i(t) = 0$.

With random access, each user transmits in a time slot with the probability p_{rand}, independent of the other users and independent of the past history. The protocol is similar to slotted ALOHA, but there is no feedback from the receiver, and the number of nodes is finite. The scheduling of the transmission of the packets is determined solely by the protocol sequence assigned to the users, or the random sequence used in the random access scheme.

We compare deterministic and random access with the performance of average delay. The *delay* or latency experienced by a user is measured as follows: starting from a random time, record the time duration a user would wait until at least one packet from each of the other $M-1$ users

Y. Wu and L. Shen are with the National Mobile Communications Research Laboratory, Southeast University, China. Email: wuyi@fjnu.edu.cn, lfshen@seu.edu.cn.

K. W. Shum is with the Institute of Network Coding, The Chinese University of Hong Kong. Email: wkshum@inc.cuhk.edu.hk.

W. S. Wong is with the Dept. of Information Engineering, The Chinese University of Hong Kong. Email: wswong@ie.cuhk.edu.hk.

Z. Lin is with the School of Electrical and Information Engineering, The University of Sydney. Email: zihuai.lin@sydney.edu.au. Z. Lin and Y. Wu are also with the Depart. of Telecomm. and Network Engineering, Fujian Normal University, China.

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would be received. An approximate formula for the distribution function of the delay in the random access scheme is derived in Appendix A.

For deterministic access with protocol sequence, we define the *duty factor* of a protocol sequence $s(t)$ as $f \triangleq \frac{1}{L}\sum_{t=0}^{L-1} s(t)$, which is nothing but the total number of ones in a period, normalized by the sequence period.

A sequence set is called *user-irrepressible* (UI) if no matter what the delay offsets are, each user can send at least one packet successfully in a period [8]. UI sequences have the favorable property that the delay is no more than one period with probability 1.

The *Hamming weight* of a sequence is the number of ones in a period. The *Hamming cross-correlation* between two sequences $a(t)$ and $b(t)$ is the number of overlapping ones between $a(t)$ and the sequence obtained by cyclically shifting $b(t)$ by some delay offset τ.

III. Constructions of Protocol Sequences

There are many methods for constructing protocol sequences in the literature See e.g. [6] In this section, We briefly review the constructions of four families of protocol sequences: PS, EPS, CRT and CRT-UI. The first two are constructed in the context of optical code-division multiple-access (OCDMA) systems. We refer the reader to [9] for their applications to OCDMA.

In the following, the parameter p is a prime number larger than or equal to the number of users M. Each of the four sequence families may consist of more than M sequences. We can assign any M of them to the M users. We define $[x \bmod n]$ to be the remainder of x after division by n, which is an integer between 0 and $n-1$.

Given a prime p, the set of *prime sequences* (PS) with parameter p consists of p sequences of length p^2, each with Hamming weight p. For $i = 0, 1, \ldots, p-1$, the i-th prime sequence, $s_i(t) = 1$ if

$$t = jp + [ij \bmod p] \tag{1}$$

for some $j = 0, 1, \ldots, p-1$, and 0 otherwise. To represent the sequence more compactly, we define the *characteristic set* as the set of the locations of "1" in the sequence. The i-th prime sequence is defined by characteristic set

$$\{jp + [ij \bmod p] : j = 0, 1, \ldots, p-1\}.$$

For example, for $p = 3$, we have the following three prime sequences:

$$\begin{aligned} s_1(t) &: 100\,100\,100, \\ s_2(t) &: 100\,010\,001, \\ s_3(t) &: 100\,001\,010. \end{aligned}$$

The corresponding characteristic sets are respectively $\{0, 3, 6\}$, $\{0, 4, 8\}$ and $\{0, 5, 7\}$.

The *extended prime sequences* (EPS) are of period $p(2p-1)$, constructed by inserting $p-1$ zeros after every "1" in a prime sequence. For example, the following three sequences are obtained after zero padding.

$$\begin{aligned} s_1(t) &: 10000\,10000\,10000, \\ s_2(t) &: 10000\,01000\,00100, \\ s_3(t) &: 10000\,00100\,01000. \end{aligned}$$

For $i = 0, 1, \ldots, p-1$, the i-th sequence in EPS has characteristic set

$$\{j(2p-1) + [ij \bmod p] : j = 0, 1, \ldots, p-1\}.$$

It can be proved that the Hamming cross-correlation of PS is between 0 and 2, while the Hamming cross-correlation of EPS is either 0 or 1 [9]. Any subset of $(p+1)/2$ prime sequences, and the whole set of p extended prime sequences are UI.

The third family is a variation of the CRT sequences presented in [10]. We also call them *CRT sequences* in this paper. Given the number of users M, let p be a prime number larger than or equal to M, and q be $p+1$.

For $g = 1, 2, \ldots, p$, the g-th sequence, $s_g(t)$ is of period pq, and $s_g(t) = 1$ if and only if t satisfies

$$g \cdot [t \bmod q] \equiv t \bmod p. \tag{2}$$

The multiplication "$\cdot$" on the left hand side of (2) is mod-p multiplication, and $[t \bmod q]$ is a integer between 0 and $q-1$. It can be shown that in each sequence so constructed, there are q ones in each period. The Hamming weight of each CRT sequence is q. It can be shown that the Hamming cross-correlation is at most two. The analysis relies on Chinese Remainder Theorem (CRT) and is similar to the proofs in [10]. The following is an example of CRT sequences for $p = 3$ and $q = 4$:

$$\begin{aligned} s_1(t) &: 111\,100\,000\,000, \\ s_2(t) &: 100\,101\,000\,010, \\ s_3(t) &: 100\,100\,100\,100. \end{aligned}$$

Like the PS, any subset of $(p+1)/2$ CRT sequences defined above is UI.

The last family is called *CRT-UI*. It is a generalization of the extended prime sequence. We take p to be a prime integer and let q be an integer larger than or equal to $2p-1$. The constructed sequences have common period pq. For $i = 0, 1, \ldots, M-1$, the i-th sequence is defined by the characteristic set

$$\{jq + [ij \bmod p] : j = 0, 1, 2, \ldots, p-1\}. \tag{3}$$

We denote this set of protocol sequences by CRT-UI(p, q).

Fig. 1 is an example generated with $p = 5, q = 10$, and the sequence period is 50.

When $q = 2p-1$, the CRT-UI and EPS are the same set of sequences.

The CRT-UI sequences have the property that (i) the period is pq, (ii) the Hamming weight is M, and (iii) the Hamming cross-correlation between two distinct sequences is either 0 or 1. One can easily deduce that CRT-UI sequences are user-irrepressible from property (ii) and (iii).

IV. Analysis of Delay Performance

In this section we consider the delay of user-irrepressible protocol sequences. We differentiate two kinds of delay measures - individual delay and group delay.

Suppose that there are K active users, and user i transmits packets according to protocol sequence $s_i(t)$, for $i = 1, 2, \ldots, K$. We start at a random time. The *individual delay* of user i is the duration user i has to wait

$$
\begin{array}{ll}
s_0(t): & 1000000000\ 1000000000\ 1000000000\ 1000000000\ 1000000000 \\
s_1(t): & 1000000000\ 0100000000\ 0010000000\ 0001000000\ 0000100000 \\
s_2(t): & 1000000000\ 0010000000\ 0000100000\ 0100000000\ 0001000000 \\
s_3(t): & 1000000000\ 0001000000\ 0100000000\ 0000100000\ 0010000000 \\
s_4(t): & 1000000000\ 0000100000\ 0001000000\ 0010000000\ 0100000000
\end{array}
$$

Fig. 1. CRT-UI sequences, $p = 5, q = 10$

after the random time until he/she can transmit successfully. We denote the individual delay of user i by random variable X_i. The individual delay X_i is a function of the delay offsets of the users. We define the group delay, denoted by Y, by the duration that the users have to wait after a random time until all of them have transmitted at least one uncollided packet The group delay can be viewed as the maximum of the individual delay $Y = \max\{X_j : j = 1, 2, \dots, K\}$.

We focus on one user, say user k, whose protocol sequence $s(t)$ has characteristic set $\mathcal{I}_k = \{x_{[1]}, x_{[2]}, \dots x_{[w]}\}$ and period L, with $x_{[1]} < x_{[2]} < \dots < x_{[w]}$. We denote the interval between the i-th and the $(i-1)$-st ones in the protocol sequence by $\delta_i \triangleq [x_{[i]} - x_{[i-1]} \bmod L]$, for $i = 2, \dots, w$ and let $\delta_1 \triangleq [x_{[1]} - x_{[w]} \bmod L]$. The vector $\boldsymbol{\delta} \triangleq (\delta_1, \delta_2, \dots, \delta_w)$ indicates how the w ones in $s(t)$ are spread between 0 and $L-1$. It is easy to check that the sum of the components in vector $\boldsymbol{\delta}$ is exactly the period L.

For example, the following protocol sequence $s_1(t)$, the characteristic set is $\{0, 4, 8\}$ and the period is $L = 15$:

$$s_1(t): \quad 1\underbrace{0001}_{\delta_2=4}\underbrace{0001}_{\delta_3=4}000000.$$

The intervals between consecutive ones are $\delta_1 = 7$, $\delta_2 = 4$, and $\delta_3 = 4$.

For $d = 0, 1, 2, \dots, w-1$, let C_d be the cyclic autocorrelation of the sequence $\boldsymbol{\delta} = (\delta_1, \delta_2, \dots, \delta_w)$,

$$C_{\boldsymbol{\delta}}(d) \triangleq \sum_{j=0}^{w-1} \delta_j \delta_{j \ominus d},$$

where the subtraction $j \ominus d$ in the subscript of δ is performed mod w.

We start with a theorem which is valid in general for sequences with Hamming cross-correlation upper bounded by 1.

Theorem 1: In the deterministic access scheme with K active users, suppose that the protocol sequences have common period L and the Hamming cross-correlation upper bounded by 1. Let the characteristic set of the i-th sequence be $\mathcal{I}_i$, for $i = 1, 2, \dots, K$, and $|\mathcal{I}_i| = w \geq K$. Consider a particular user, say user k. Let the intervals between adjacent ones in the protocol sequence of user k be recorded in the vector $\boldsymbol{\delta} = (\delta_1, \dots, \delta_w)$. If the relative delay offsets of each user is uniformly and independently distributed between 0 and $L-1$, then the expected value of the individual delay is

$$\mathbb{E}[X_k] = \frac{1}{L}\Big[\frac{1}{2}C_{\boldsymbol{\delta}}(0) + \sum_{n=1}^{K-1}(1-p_n)C_{\boldsymbol{\delta}}(n)\Big] - \frac{1}{2}. \qquad (4)$$

where

$$p_n \triangleq \sum_{i=1}^{n}(-1)^{i+1}\binom{n}{i}\Big(1 - \frac{iw}{L}\Big)^{K-1} \qquad (5)$$

is defined for $n = 1, 2, \dots, w$.

Proof: We first note that the sequence set is user-irrepressible, because the Hamming weight w is no less than the number of active users, and the Hamming cross-correlation is no more than 1.

Suppose that the relative delay offset of user k is fixed at τ. Let $x_1(\tau)$, $x_2(\tau), \dots, x_w(\tau)$ be integers sorted in ascending order such that $\mathcal{I}_k + \tau = \{x_1(\tau), x_2(\tau), \dots, x_w(\tau)\}$. Suppose that x is between $x_n(\tau)$ and $x_{n+1}(\tau)$. User k can transmit a packet successfully before time x if and only if it can transmit at least one successful packet at time $x_1(\tau)$, $x_2(\tau), \dots$, or $x_n(\tau)$. With the assumption that the Hamming cross-correlation is no more than 1, another user can collide with user k only once among the packets transmitted at time $x_1(\tau)$, $x_2(\tau), \dots$, or $x_n(\tau)$. We can apply the principle of inclusion-and-exclusion and conclude that this probability is equal to the probability p_n defined in (5). The conditional cumulative distribution function (cdf) of individual delay X_k is thus given by

$$\Pr(X_k \leq x|\tau) = \begin{cases} 0 & \text{for } 0 \leq x < x_1(\tau) \\ p_n & \text{for } x_n(\tau) \leq x < x_{n+1}(\tau),\ n = 1, \dots, w. \end{cases}$$

We remark that the probability p_n is equal to 1 for $K \leq n \leq w$. It conforms with the fact that user j cannot be blocked by the others, i.e., $P(X_k \leq x_K(\tau)|\tau) = 1$.

The conditional expected value of X_k can be computed by

$$\mathbb{E}[X_k|\tau] = \sum_{t=1}^{\infty}\Pr(X_k \geq t|\tau) = \sum_{t=1}^{L-1}\Pr(X_k \geq t|\tau).$$

The infinite summation can be replaced by a finite one because $\Pr(X_k \geq t|\tau) = 0$ for $t \geq L$. Hence,

$$\mathbb{E}[X_k|\tau] = \sum_{t=0}^{L-1}\Pr(X_k > t|\tau) = \sum_{t=0}^{L-1}\big(1 - \Pr(X_k \leq t|\tau)\big). \qquad (6)$$

In the second summation in (6), the first $x_1(\tau)$ terms correspond to $t = 0, 1, \dots, x_1(\tau) - 1$, and are all equal to 1. The next $x_2(\tau) - x_1(\tau)$ terms correspond to $t = x_1(\tau), x_1(\tau)+1, \ldots, x_2(\tau)-1$, and are equal to $1-p_1$, and so on. We can group the like terms together and obtain

$$\mathbb{E}[X_k|\tau] = x_1(\tau) + \sum_{n=1}^{K-1}(1-p_n)(x_{n+1}(\tau) - x_n(\tau)). \quad (7)$$

We only need to sum up to $n = K-1$, because $p_n = 1$ for $n \geq K$.

Summing (7) over all τ, we get

$$\begin{aligned}\sum_{\tau=0}^{L-1}\mathbb{E}[X_k|\tau] &= \sum_{\tau=0}^{L-1}\Big[x_1(\tau) + \sum_{n=1}^{K-1}(1-p_n)(x_{n+1}(\tau) - x_n(\tau))\Big] \\ &= \sum_{\tau=0}^{L-1} x_1(\tau) + \sum_{n=1}^{K-1}(1-p_n)\sum_{\tau=0}^{L-1}(x_{n+1}(\tau) - x_n(\tau)).\end{aligned}$$

The summation $\sum_{\tau=0}^{L-1} x_1(\tau)$ is the sum of the minimum elements in $\mathcal{I}_{\|} + \tau$, and is equal to

$$\sum_{j=1}^{w}\sum_{k=0}^{\delta_j-1} k = \sum_{j=1}^{w}\frac{\delta_j(\delta_j-1)}{2} = \frac{C_{\boldsymbol{\delta}}(0) - L}{2}.$$

Let δ_j be the j-th component in the vector $\boldsymbol{\delta}$ of inter-packet intervals. The interval δ_j appears as a summand in

$$\sum_{\tau=0}^{L-1}(x_{n+1}(\tau) - x_n(\tau))$$

exactly $\delta_{j\ominus n}$ times. Hence,

$$\sum_{\tau=0}^{L-1}(x_{n+1}(\tau) - x_n(\tau)) = \sum_{j=1}^{w}\delta_j\delta_{j\ominus n} \triangleq C_{\boldsymbol{\delta}}(n).$$

Consequently, we obtain

$$\mathbb{E}[X_k] = \frac{1}{L}\sum_{\tau=0}^{L-1}\mathbb{E}[X_k|\tau] = \frac{1}{L}\Big[\frac{C_{\boldsymbol{\delta}}(0)-L}{2} + \sum_{n=1}^{K-1}(1-p_n)C_{\boldsymbol{\delta}}(n)\Big],$$

which can be simplified to (4). ■

Corollary 2: With the assumption and notation as in Theorem 1, we have

$$\mathbb{E}[X_k] \leq \frac{C_{\boldsymbol{\delta}}(0)}{L}\Big[\frac{1}{2} + \sum_{n=1}^{K-1}(1-p_n)\Big] - \frac{1}{2}.$$

Proof: From Theorem 1, it is clear that

$$\mathbb{E}[X_k] \leq \frac{\max_{0\leq n<K} C_{\boldsymbol{\delta}}(n)}{L}\Big[\frac{1}{2} + \sum_{n=1}^{K-1}(1-p_n)\Big] - \frac{1}{2}.$$

However, by Cauchy-Schwarz inequality, we have

$$C_{\boldsymbol{\delta}}(n)^2 = \Big(\sum_{j=1}^{w}\delta_j\delta_{j\ominus n}\Big)^2 \leq \Big(\sum_{j=1}^{w}\delta_j^2\Big)\Big(\sum_{j=1}^{w}\delta_{j\ominus n}^2\Big) = C_{\boldsymbol{\delta}}(0)^2.$$

The result follows by replacing $\max_n C_{\boldsymbol{\delta}}(n)$ by $C_{\boldsymbol{\delta}}(0)$. ■

In view of the above upper bound of $\mathbb{E}[X_k]$, a protocol sequence with $C_{\boldsymbol{\delta}}(0)$ as small as possible is desirable, as far as the individual delay is concerned. The value of $C_{\boldsymbol{\delta}}(0)$ is the sum of the squares of inter-packet times, and is minimized if the ones in the protocol sequence are evenly spread. For protocol sequence with period L and the Hamming weight w, the smallest possible value of $C_{\boldsymbol{\delta}}(0)$ is equal to $w(L/w)^2 = L^2/w$.

We henceforth call a protocol sequence *quasi-random* if $C_{\boldsymbol{\delta}}(0)$ is close to the minimum value. It measures the "evenness" of the distribution of the ones in a protocol sequence. The optimal value is $C_{\boldsymbol{\delta}}(0) = w(L/2)^2$, and is achievable only if the period L is divisible by w and the separation between two consecutive ones in the sequence is equal to L/w.

Theorem 3: For $q \geq 2p-1$, a sequence from CRT-UI(p,q) has $C_{\boldsymbol{\delta}}(0)$ upper bounded by $\Big(1 + \frac{p^2}{4q^2}\Big)pq^2$.

Proof: Let $s_g(t)$, $g = 0, 1, \ldots, p-1$ be the protocol sequences in CRT-UI(p,q). In the first sequence $s_0(t)$, the distance between two consecutive ones is exactly q. Therefore for $s_0(t)$, $C_{\boldsymbol{\delta}}(0) = pq^2 = L^2/p$. The first sequence in CRT-UI(p,q) achieves the minimum possible value for $C_{\boldsymbol{\delta}}(0)$.

Consider the sequence $s_g(t)$ generated by g, for $g = 1, 2, \ldots, p-1$. Recall that the characteristic set of $s_g(t)$ is defined as $\{[gn \bmod p] + nq : n = 0, 1, \ldots, p-1\}$, where $[gn \bmod p]$ is the remainder of gn after division by p. The difference between the n-th packet and the $(n+1)$-st packet is $\delta_{n+1} = q + [g(n+1) \bmod p] - [gn \bmod p]$. Because $[g(n+1) \bmod p]$ is equal to

$$\begin{cases}[gn \bmod p] + g & \text{if } 0 \leq [gn \bmod p] < p-g, \\ [gn \bmod p] + g - p & \text{if } p-g \leq [gn \bmod p] < p-1\end{cases}$$

the value $q+g$ appears $p-g$ times as a component in the vector $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_p)$, and the value $q+g-p$ appears g times. Hence, for $s_g(t)$, we have $C_{\boldsymbol{\delta}}(0) = (q+g)^2(p-g) + (q+g-p)^2(g)$. After some simplifications, we can write

$$C_{\boldsymbol{\delta}}(0) - pq^2 = gp(p-g).$$

We treat the right-hand side as a quadratic function in g, with g being a continuous variable between 0 and p. The maximum of $gp(p-g)$ is attained at $g = p/2$, with maximal value $p^3/4$. Therefore $C_{\boldsymbol{\delta}}(0) - pq^2 \leq p^3/4$. This proves that $C_{\boldsymbol{\delta}}(0) \leq (1 + p^2/(4q^2))pq^2$. ■

Since $q \geq 2p-1$, we can upper bound $p^2/(4q^2)$ approximately by $1/16$. We thus have the following approximate upper bound for the expected individual delay for CRT-UI(p,q), with $q \geq 2p-1$,

$$\mathbb{E}[X_k] \leq \frac{17Lf}{16}\Big[\frac{1}{2} + \sum_{n=1}^{K-1}(1-p_n)\Big] - \frac{1}{2}, \quad (8)$$

where $L = pq$ is the sequence period, $f = 1/q$ is the duty factor, and p_n is given in (5).

In Fig. 2, we plot the upper bound in (8) for a system of K active users with CRT-UI$(p, 2p-1)$, where p is a prime larger or equal to p. We compare with random schemes with duty factors equal to $1/K$ and $1/(2K-1)$. The duty factor of the scheme with CRT-UI$(p, 2p-1)$ is roughly the same as the random scheme with duty factor $1/(2K-1)$. The difference between these two is big. However, duty factor $1/(2K-1)$ is not the optimal choice that minimizes the expected individual delay. The optimal duty factor turns out to be $1/K$. From Fig. 2, we see that the deterministic scheme is still better than the random scheme with duty factor $1/K$. As the actual individual delay is smaller than the upper bound, the actual performance gap is larger than the gap we can see in Fig. 2.

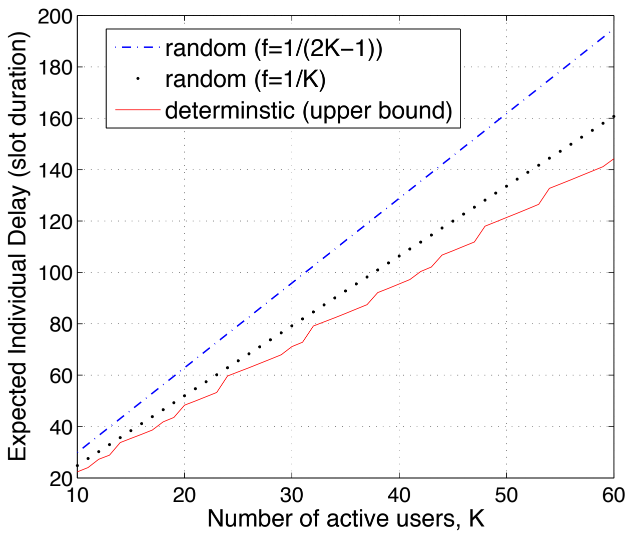

Fig. 2. Cumulative distribution functions of delay for 31 users, no transmission error.

V. Deterministic vs random access scheme

We compare the performance of deterministic and random access scheme in the scenario that the system is fully loaded, that is, all sequences in the sequence set are assigned to the users and the users are all active.We further assume that the channel introduces packet erasures, that is, if one user gets a transmission opportunity at a time slot, the packet can be correctly received by users in listening mode with a probability of $(1-\epsilon)$, for some constant ϵ between 0 and 1. Hence, a packet can be received successfully when the sender is the only sender in a time slot, and the packet is not erased by channel impairment.

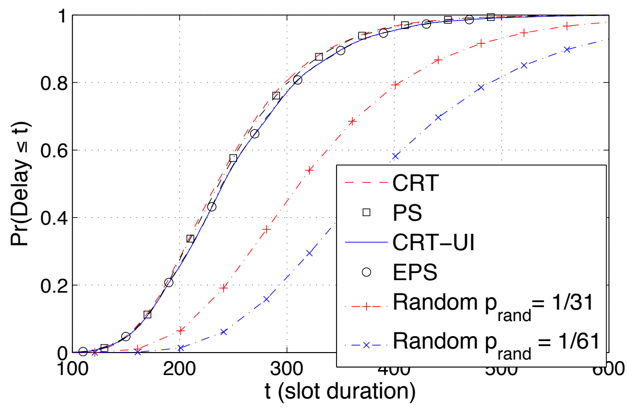

Fig. 3. Cumulative distribution functions of delay for 31 users, no transmission error.

We first consider a system in which every user is active and there is no error on each transmission link, the packets will be dropped only due to collision. For $M = 31$ users, the cumulative distribution function (CDF) pertaining to CRT sequences, PS , CRT-UI sequences, and EPS, all with parameter p set to 31, are shown in Fig. 3. The duty factors of these four sequence sets are respectively 1/31, 1/31, 1/61 and 1/61. In random access scheme, the optimal probability of transmitting a packet in a time slot is $p_{rand} = 1/31$. We also plot the CDF of random access with transmitting probability $p_{rand} = 1/61$, in order to compare with EPS and CRT-UI. We can see a significant gap between the CDFs of the deterministic schemes and the CDFs of the random access scheme. One can see (from the color version of Fig. 3), that the delay with CRT sequences is slightly better than that with PS, but the difference is very small and negligible. The delay with EPS is virtually the same if CRT-UI is used.

Since blocking is impossible for CRT-UI and EPS, which are known to be UI, we have $\Pr(\text{delay} \le p(2p-1)) = 1$ for CRT-UI and EPS. However, blocking may happen with a very small probability for CRT and PS. Hence $\Pr(\text{delay} < \infty) < 1$ for CRT and PS. For random access scheme, we have $\Pr(\text{delay} \le t) \to \infty$ as $t \to \infty$.

To explain the gain achieved by deterministic scheme, we can consider a simple case in which *only two* users are active, and both of them are using protocol sequences with Hamming cross-correlation no more than one, such as CRT-UI or EPS. It implies that there cannot be two consecutive collisions within a period; if a packet is involved in collision, it is guaranteed that the next packet will not collide. Hence, consecutive collisions are not possible. However if random sequence is used as the protocol sequence, consecutive collisions are possible.

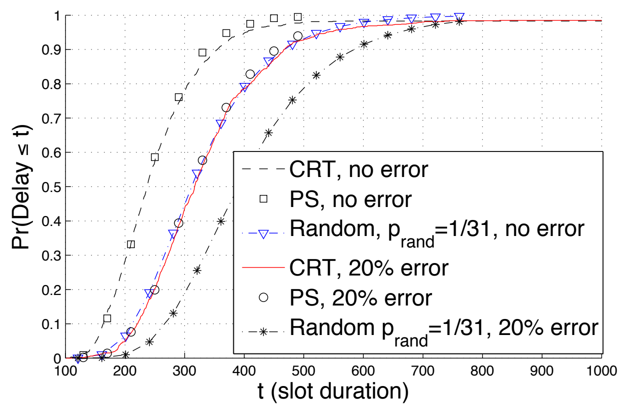

Fig. 4. Cumulative distribution functions of delay for 31 users for CRT, PS and random p_{rand}=1/31 sequences, the transmission error probability is 20 percent.

Fig. 3 presented the results for wireless networks in which no errors exist in the transmission link. All the packets can be correctly received by the receiver if there is no collisions among different packets. For a real wireless network, this is usually an invalid assumption due to the wireless channel transmission property. Fading, interference and noise effects make the wireless channel transmission highly unreliable. Fig. 4 and Fig. 5 show the CDF for $M = 31$ users with channel transmission error probabilities of 0.2 for CRT, PS and CRT-UI, EPS sequences, respectively. From these figures, one can see that with the channel transmission errors, the deterministic schemes still

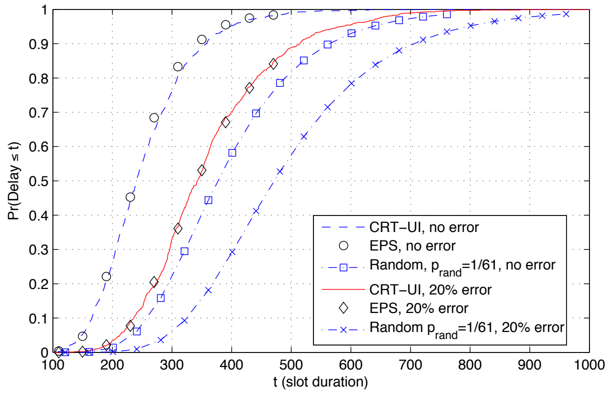

Fig. 5. Cumulative distribution functions of delay for 31 users or CRT-UI, EPS and random p_{rand}=1/61 sequences, the transmission error probability is 20 percent.

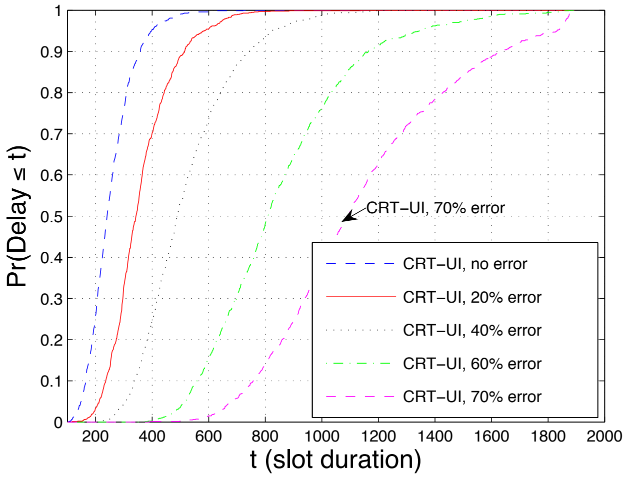

Fig. 6. Cumulative distribution functions of delay for 31 users for CRT-UI sequence with different transmission error probability.

perform better than the corresponding random schemes. As the channel transmission error probability increases, the delay enlarges for all the investigated schemes. The influence of channel transmission error probability to the delay for CRT-UI sequences can be seen from Fig. 6, where the CDF curves for $M = 31$ users with different channel transmission error probabilities are shown. It can be seen that with CRT-UI sequences, 50% users will suffer from more than 800 slots delay when the transmission error increases from 0 to 70%.

VI. Concluding Remarks

Several constructions of protocol sequences are described and compared with average delay. They are statistically different from random sequences. The structures and patterns in the deterministic protocol sequences are exploited to yield better performance than the random approach. From the simulation, we see that the PS and CRT are close in performance. The performance of EPS and CRT-UI are also very close. But the CRT sequences considered in this paper is just a special case of the general construction in [10]. The construction of CRT-UI is more flexible then EPS. The two CRT constructions provide more design options.

Appendix

I. Delay in Random Access Scheme

Suppose that the M users start transmitting at time zero. Each of them transmits at a time slot with probability p_{rand}, independent of the others. We focus on one user, say user k. For $j \neq k$, let X_{jk} be the time when user k first receives a packet from user j. The random variable X_{jk} is geometrically distributed.

$$Pr(X_{jk} \leq i) = 1 - \beta^i, i = 1, 2, 3, \ldots$$

where $\beta = 1 - p_{rand}(1 - p_{rand})^{M-1}$ is the probability that user j either remains silent in a time slot or transmits a packet with collision.

Further, if the packet can be correctly received by users in listening mode with a probability of $(1 - \epsilon)$ in noisy channel. Then, β can be replaced with $1-p_{rand}(1-\epsilon)(1-p_{rand})^{M-1}$.

Let $X_k = \max_{j\neq k} X_{jk}$ denote the time that user k has to wait until at least one packet from each of the other $M-1$ users is received. The distribution function of X_k is

$$\Pr(X_k \leq i) = \prod_{j\neq k} \Pr(X_{jk} \leq i) = (1 - \beta^i)^{M-1}$$

where the product is over all j from 1 to M except k. The expected delay experienced by user k is

$$\mathbb{E}[X_k] = \sum_{i=0}^{\infty} \Pr(X_k > i) = \sum_{i=0}^{\infty} [1 - (1 - \beta^i)^{M-1}]. \quad (9)$$

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